

CH 2: BAYESIAN DECISION THEORY

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Introduction to Bayesian Decision Theory

Bayesian Decision Theory is a **probabilistic approach** used for decision-making under **uncertainty**. It relies on **Bayes' Theorem** to update probabilities based on new information. It quantifies the trade-offs between the costs that accompany such classifications

Assumptions:

- Decision problem is posed in probabilistic terms
- All relevant probability values are known

Concept of Probability in Decision Making

In pattern recognition and classification problems, decisions must be made based on **uncertain or incomplete information**. Probability provides a mathematically sound way to handle this uncertainty. **Types of Probability in Decision Making:**

1. Prior Probability ($P(\omega)$):

- The initial belief about the probability of a class (ω) before any observation.
- For example, if you know that 30% of the emails are spam, then $P(\text{spam}) = 0.3$.

2. Likelihood ($P(\mathbf{x}|\omega)$):

- The probability of observing a particular feature vector $\mathbf{x}$, given that the actual class is ω .
- This is derived from the class-conditional density function.

3. Posterior Probability ($P(\omega|\mathbf{x})$):

- The updated probability of the class ω after observing $\mathbf{x}$.
- It represents our confidence that the observation $\mathbf{x}$ belongs to class ω .

4. Evidence or Marginal Likelihood ($P(\mathbf{x})$):

- The total probability of observing $\mathbf{x}$ across all classes.
- Used as a normalizing constant in Bayes' Theorem.

Bayes' Theorem and Conditional Probability

Bayes' Theorem is used to update our beliefs (posterior probabilities) based on new evidence. It is fundamental in Bayesian Decision Theory.

Bayes' Theorem:

$$P(\omega_i|x) = \frac{P(x|\omega_i) \cdot P(\omega_i)}{P(x)}$$

- $P(\omega_i|x)$: Posterior probability of class ω_i given feature vector x
- $P(x|\omega_i)$: Likelihood of x given class ω_i
- $P(\omega_i)$: Prior probability of class ω_i
- $P(x)$: Evidence or total probability of x

Conditional Probability:

The probability of an event **A**, given that another event **B** has occurred, is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

In the context of pattern recognition:

- A = instance belongs to class ω_i
- B = observation or measurement x

Classifiers in Bayesian Decision Theory

Bayesian Decision Theory uses **probability models** to perform classification. It aims to assign an observation or feature vector to the most probable class based on available data. The classifiers under this framework are designed to make **optimal decisions under uncertainty**.

Bayes Classifier

The Bayes classifier is a **theoretical optimal classifier** that minimizes the **probability of classification error**. It uses **posterior probabilities** calculated via Bayes' theorem and assigns a sample to the class with the **highest posterior**.

Decision Rule:

Given an input vector x , the Bayes classifier assigns it to the class ω_i such that:

$$\omega_{\text{assigned}} = \arg \max_{\omega_i} P(\omega_i|x)$$

[arg max finds the argument(ω_i) for which the probability is maximum]

Using Bayes' theorem:

$$P(\omega_i|x) = \frac{P(x|\omega_i) \cdot P(\omega_i)}{P(x)}$$

Since $P(x)$ is the same for all classes, the decision depends on:

$$\omega_{\text{assigned}} = \arg \max_{\omega_i} P(x|\omega_i) \cdot P(\omega_i)$$

Characteristics:

- Requires knowledge of class priors $P(\omega_i)$ and likelihoods $P(x|\omega_i)$
- Minimizes classification error (optimal in theory)
- Not always practical due to the need for **complete probability distributions**

Naïve Bayes Classifier

The Naïve Bayes classifier is a **simplified version of the Bayes classifier**. It assumes that all features are **conditionally independent** given the class label.

Formula:

If $x = (x_1, x_2, \dots, x_n)$ is the feature vector, then:

$$P(x|\omega_i) = \prod_{j=1}^n P(x_j|\omega_i)$$

So, the decision rule becomes:

$$\omega_{\text{assigned}} = \arg \max_{\omega_i} P(\omega_i) \cdot \prod_{j=1}^n P(x_j|\omega_i)$$

Characteristics:

- Easy to implement
- Performs well in many real-world applications despite the independence assumption
- Requires fewer parameters than the full Bayes classifier
- Often used in text classification, spam detection, sentiment analysis

Bayesian Networks (Bayes Nets)

A Bayesian Network is a **graphical model** that represents a set of variables and their conditional dependencies via a **directed acyclic graph (DAG)**.

Each node in the graph represents a random variable (feature or class), and each edge represents a probabilistic dependency.

Properties:

- Captures **conditional dependencies** more flexibly than Naïve Bayes
- The joint probability distribution over all variables is factored as:

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(x_i))$$

Where **Parents(x_i)** is the set of nodes that have directed edges pointing to x_i.

Advantages:

- Can model more complex dependencies than Naïve Bayes
- Allows inference, reasoning, and learning from data
- Useful in diagnostic systems, decision support, and causal modelling.

Discriminant Functions

A **discriminant function** is a mathematical function that takes a feature vector x as input and produces a value that helps decide **which class** the input belongs to.

A **discriminant function** $g_i(x)$ is associated with each class ω_i . For a given input x , we compute all $g_i(x)$, and assign x to the class ω_i for which $g_i(x)$ is the **largest**.

Assign x to class ω_i if $g_i(x) > g_j(x)$ for all $j \neq i$

Role in Classification:

- Discriminant functions simplify decision-making.
- Instead of computing and comparing probabilities directly, we use functions that combine prior probabilities, likelihoods, or features in a **compact form**.
- They're especially useful in **multi-class classification** problems.

Linear Discriminant Functions

A **linear discriminant function** has the form:

$$g_i(x) = w_i^T x + w_{i0}$$

- w_i is a **weight vector** for class ω_i
- x is the input **feature vector**
- w_{i0} is a **bias term** (scalar)
- $w_i^T x$ is the **dot product** of the vectors

This equation defines a **hyperplane** that separates the feature space into different regions — each corresponding to one class.

Example: If there are two classes ω_1 and ω_2 , and the decision boundary is a line (in 2D), a linear function like:

$$g(x) = w^T x + w_0 = 0$$

defines the decision boundary. If $g(x) > 0$, assign to one class; if $g(x) < 0$, assign to the other.

Characteristics:

- Decision surfaces are straight lines (2D), planes (3D), or hyperplanes (nD).
- Fast and simple
- Work well when classes are **linearly separable**

Non-Linear Discriminant Functions

These functions can define **curved** or more complex boundaries between classes. Example forms:

$$g_i(x) = x^T A_i x + b_i^T x + c_i$$

- A_i is a matrix defining **quadratic terms**
- b_i is a vector
- c_i is a scalar

This is a **quadratic discriminant function**. It can separate classes that are **not linearly separable**.

Characteristics:

- Can model complex, non-linear decision boundaries
- Used in **Quadratic Discriminant Analysis (QDA)** and **Kernel methods**
- More flexible, but computationally more expensive

Type	Equation Form	Decision Surface	When to Use
Linear	$g_i(x) = w_i^T x + w_{i0}$	Line / Plane / Hyperplane	Classes are linearly separable
Non-Linear	$g_i(x) = x^T A_i x + b_i^T x + c_i$	Curve / Complex Surface	Classes overlap or are not linearly separable

Decision Surfaces

A **decision surface** (also called a decision boundary) is a geometric boundary in the feature space that **separates different classes**. It is formed by the set of points x for which the **discriminant functions of two or more classes are equal**.

Mathematically, the decision surface between class ω_i and class ω_j is defined as:

$$g_i(x) = g_j(x)$$

This surface **divides the feature space** into regions, each corresponding to a different class. When a new sample is given, the classifier checks **which region the sample lies in**, and assigns it to the corresponding class.

Importance:

- The decision surface determines **how effectively a classifier can separate classes**.
- Its **shape and complexity** affect both classification **accuracy** and **computational cost**.
- Understanding decision surfaces helps in selecting appropriate classifiers (linear vs non-linear).
- In real-world problems, especially with overlapping classes, the nature of decision surfaces plays a critical role in performance.

Types of Decision Surfaces

Type	Linear Decision Surface	Quadratic Decision Surface	Non-Linear Decision Surface
Formed by	Linear discriminant functions $g(x) = w^T x + w_0$	Quadratic discriminant functions $g(x) = x^T A x + b^T x + c$	Non-linear functions, often using techniques like: • Kernel methods (in SVMs) • Neural networks • Decision trees
Surface type	Hyperplane in n-dimensional space	Parabolas, ellipses, hyperbolas	Arbitrary, complex shapes
Properties	• Separates data using straight lines	• data where classes are not linearly separable	• Can model highly complex decision boundaries

	(2D), planes (3D), or hyperplanes - Fast to compute and easy to interpret - Works well when data is linearly separable	- Decision boundaries are curved - More complex, requires more parameters	- Suitable for problems with overlapping or irregular class distributions - Requires more data and computational power
Example	- Linear Discriminant Analysis (LDA) - Perceptron - Support Vector Machines (Linear Kernel)	- Quadratic Discriminant Analysis (QDA) - Bayesian Classifiers with Unequal Covariance Matrices	- Kernel Support Vector Machines (SVM with RBF or polynomial kernel) - Neural Networks (Multi-layer Perceptrons, CNNs) - k-Nearest Neighbors (k-NN) - Decision Trees and Random Forests
When to Use	Linearly separable classes	Non-linear but smooth boundaries	Complex class distributions

Normal Density and Discriminant Functions

Probability Density Functions (PDF)

A **probability density function (PDF)** is used to describe the likelihood of a continuous random variable falling within a particular range of values.

For a continuous feature vector x , the PDF $p(x)$ satisfies:

- $p(x) \geq 0$
- $\int p(x) dx = 1$

In classification, the class-conditional density $p(x|\omega_i)$ gives the **likelihood of observing x** given that it came from class ω_i .

Gaussian Distribution and Its Role in Classification

The **Gaussian (Normal) distribution** is a widely used probability distribution for modeling real-valued data. It's defined by:

- Mean vector μ**
- Covariance matrix Σ**

Multivariate Gaussian PDF:

$$f(x; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^k * |\Sigma|}} * e^{\left[-\frac{1}{2} * ((x-\mu)^T \cdot \Sigma^{-1} \cdot (x-\mu))\right]}$$

- x : input feature vector of dimension k
- μ : mean vector of class ω_i
- Σ_i : covariance matrix of class ω_i

Role in Classification:

- If class-conditional densities are assumed to be **Gaussian**, then Bayes' rule can be applied using these densities.
- When all classes are modeled using Gaussian PDFs, we can derive **discriminant functions** from them.

Likelihood Functions and Posterior Probabilities

Likelihood Function: For classification, the **likelihood** is: $p(x|\omega_i)$

This tells us **how likely** it is to observe a sample x , given that it comes from class ω_i .

Posterior Probability (Bayes' Rule):

$$P(\omega_i|x) = \frac{p(x|\omega_i)P(\omega_i)}{p(x)}$$

Using These in Discriminant Functions:

The goal is to compute **posterior probabilities** and pick the class with the highest value. Since the denominator $p(x)p(x)p(x)$ is constant for all classes, we usually compare:

$$g_i(x) = \ln p(x|\omega_i) + \ln P(\omega_i)$$

This is the **log-likelihood plus log prior**, forming a discriminant function.

Fish Classification Problem (Example)

The **Fish Classification Problem** is a common example used to illustrate classification techniques.

Problem Statement: Given a dataset of **fish species**, the goal is to classify a new fish into one of the predefined species based on **certain features**.

1. Features Used for Classification:

- is an observable variable.
- Is a set from which we can sample or observe values.
- Eg: Length (size of the fish), Weight, Color, Fins Shape, Scale Texture, Habitat (Freshwater/Saltwater)

2. Steps in Pattern Recognition for Fish Classification

1. **Data Collection:** Gather fish images and measurement data.
2. **Preprocessing:** Normalize size variations, remove noise, and standardize features.
3. **Feature Extraction:** Identify distinguishing characteristics like **body shape, fin length, and scale pattern**.
4. **Classification Algorithm**
 - **Supervised Learning:**
 - **k-NN:** Classify based on nearest neighbors.
 - **SVM:** Find an optimal decision boundary between species.
 - **Neural Networks:** Learn complex patterns in fish images.
 - **Unsupervised Learning:**
 - **Clustering (K-Means, Hierarchical)** to group fish species without predefined labels.
5. **Evaluation & Decision Making**
 - Use a **confusion matrix** to measure accuracy.
 - Fine-tune the model for better classification results.

- Let us assume that any given fish is either a salmon or a sea bass
- call this the **state of nature of the fish**.
- Let's define a (probabilistic) variable ω

$\omega = \omega_1$ for sea bass



Salmon

$\omega = \omega_2$ for salmon



Sea Bass

Decide ω_1 if $P(\omega_1) > P(\omega_2)$; otherwise decide ω_2 .

Key Probability Concepts in Decision Making:

1. **Prior Probability (P(Class))** → The initial probability of a class before observing data.
2. **Likelihood (P(Data | Class))** → The probability of data given a particular class.
3. **Posterior Probability (P(Class | Data))** → The updated probability of a class after observing data, calculated using Bayes' Theorem.

Example:

- Suppose **90% of emails are non-spam** ($P(\text{Non-Spam}) = 0.9$) and **10% are spam** ($P(\text{Spam}) = 0.1$).
- If a keyword "**lottery**" appears in 50% of spam emails and only 2% of non-spam emails, Bayes' Theorem helps determine whether an email containing "lottery" is **more likely to be spam or not**.

Posterior probability is the probability of a certain state of nature given our observables: $P(\omega|x)$.

$$P(w|x) = \frac{p(x|w)P(w)}{p(x)} \quad \text{Using } P(w,x) = P(w|x)p(x) = p(x|w)P(w)$$

$$= \frac{p(x|w)P(w)}{\sum_i p(x|w_i)P(w_i)}$$

LOSS FUNCTIONS

- A loss function states exactly how costly each action is.
- As earlier, we have c classes $\{\omega_1, \dots, \omega_c\}$.
- We also have a possible actions $\{\alpha_1, \dots, \alpha_a\}$.
- The loss function $\lambda(\alpha_i | \omega_j)$ is the loss incurred for taking action α_i when the class is ω_j .